

Mzuzu University
Faculty of Environmental Sciences
Department of Agri-Sciences

Introduction to Statistical Analysis
STAT 2301

END OF SEMESTER EXAMINATION

Date: 30 August, 2021

Time: **14:00-17:00**

Answer all questions

Question 1

What are basic principles of experimental designs?

25 marks

Question 2

An experiment to determine the effect of 5 treatments (A, B, C, D, E) was carried out using a Complete Randomized Design and the results are given in the table below. Conduct an Analysis of Variance.

25 marks

		Treatment				
		A	B	C	D	E
Replicate	1	42.2	28.4	18.8	41.5	33.0
	2	34.9	28.0	19.5	36.3	26
	3	29.7	22.8	13.1	31.7	30.6
	4	30	18.5	10.1	31	25
	5	41	19.4	14	28.2	22

Question 3

Descriptive statistics are used for organizing and summarizing data. Discuss 25marks

Question 4

Calculate the following statistics for **each class** given below:

- | | | |
|------|---------------------|---------|
| i. | Mean | 3 marks |
| ii. | Median | 3 marks |
| iii. | Mode | 3 marks |
| iv. | Range | 3 marks |
| v. | Interquartile range | 4 marks |
| vi. | Variance | 6 marks |
| vii. | Standard Deviation | 3 marks |

Class A	Class B
102	87
128	109
98	111
140	103
98	162
110	109
115	120
109	93
106	80
89	96
119	131
98	127
56	105